

MANASH SWAIN

AI Automation Engineer | Full-Stack Developer

swainm099@gmail.com | [LinkedIn](#) | [Github](#)

SKILLS

- **Languages:** C++, JavaScript, Python, SQL
- **Frontend:** React, Tailwind CSS, HTML5, CSS3
- **Backend & Databases:** Node.js, Express.js, FastAPI, MongoDB, ChromaDB, REST API
- **AI / LLM Engineering:** Anthropic/Claude API, Tool Use (Function Calling), Prompt Engineering, Structured Outputs (Pydantic), Agentic Workflow Design, Guardrails & Rule-Based Gating, Vector Embeddings
- **Data Analytics:** Pandas, NumPy, Matplotlib, Seaborn, SQL (joins, CTEs, window functions), Precision/Recall Evaluation
- **CS Fundamentals & Tools:** Data Structures, Algorithms, Git & GitHub
- **Coding Practice:** 300+ problems solved on LeetCode & GeeksforGeeks

PROJECTS

Gatekeeper : Bounded AI Agent for Revenue Recovery *Python, FastAPI, Claude API, SQLite, React*

[GitHub](#) | [Live Demo](#)

- **Designed and built** an LLM-driven agent that diagnoses failed payments and subscription lapses and proposes a recovery action (retry, notify, escalate, or stop), built for the Razorpay AI Intern Buildathon.
- **Architected the system** as a routing/evaluator workflow rather than a fully autonomous agent, so every AI-proposed action passes through a deterministic rules layer before executing the model never has final authority over a money action.
- **Implemented context-aware reasoning** via tool-use API, retrieving customer retry history and payment-method status before generating a structured, Pydantic-validated decision.
- **Ran and evaluated the agent** across a 100+ record synthetic dataset, computing recovery totals, false-positive rate and cost, and rule-override frequency with a full audit trail logging every proposal, override, and outcome.
- **Engineered graceful failure handling** by simulating a tool-call failure and having the agent default to safe escalation instead of failing silently, demonstrated in a live dashboard.

TaskFlow Full-Stack Social Task Management App (MERN) *React, Node.js, Express, MongoDB*

[GitHub](#) | [Live Demo](#)

- **Built a full-stack task manager** with JWT authentication, bcrypt password hashing, and protected routes enforced on both frontend and backend.
- **Implemented** task CRUD, subtask tracking, and sorting/filtering by priority, category, and status.
- **Designed a dashboard** with live task statistics, profile management, and cloud image uploads via ImageKit.
- **Deployed** the client on Vercel and the API on Render, backed by MongoDB Atlas.

EDUCATION

BITS Pilani — B.S. in Computer Science 2024 – 2028

PW IOI — Skills Programme 2024 – 2028